Weasel: Out-of-Domain Generalization for Web Agents via Importance-Diversity Data Selection

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Paper version: arXiv 2605.20291v2 · CUA Radar Reel poster, not made or endorsed by the authors

Problem

> *agents fine-tuned on specific trajectories and domain often struggle to generalize out of domain*

Offline web data is **noisy and redundant**; long AXTree states make training **compute-hungry** and OOD-fragile.

Motivation: three open gaps

1. **OOD drop.** Unseen sites break in-benchmark agents.
2. **Redundant data.** Long traces, sparse task signal.
3. **Huge states.** One AXTree state = 180K tokens.

Evaluation setup

WebArena-Lite · WebArena · MiniWob · WorkArena L1/L2 · AgentTrek 52K train · Qwen2.5-7B-Instruct · Gemma3-4B-IT · Qwen3-8B

- **All online SR** in AgentLab; judges GPT-4.1-mini / GPT-4-1106-preview.
- **Held out:** zero-shot from AgentTrek / NNetNav; no eval-set tuning.

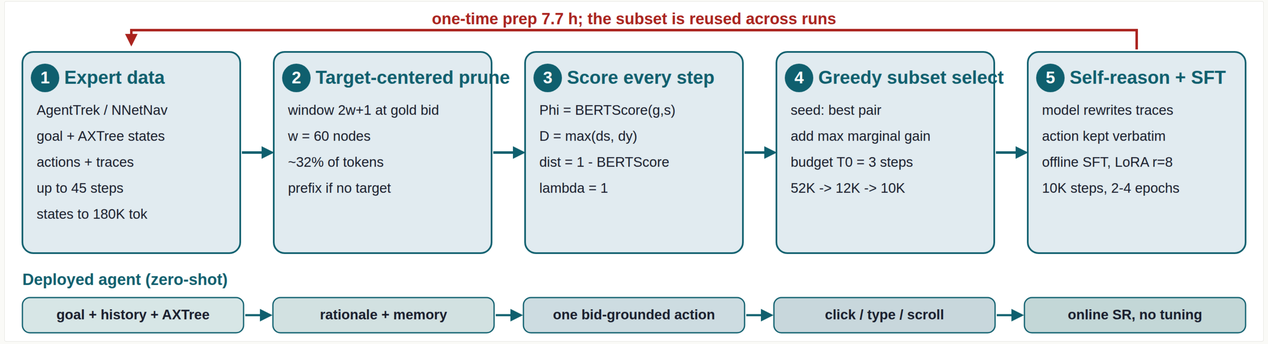
Our read (CUA Radar)

63	★★	transfer
score of 100	band	evidence level

- **Reports:** online eval; held-out envs; matched budget; cost; ablation; code.
- **Flags:** no seeds/variance; eval judge unvalidated; style-mismatch risk.
- Pattern K25. Authors' numbers, not rerun.

Method: curate the data, then train offline

- **Sees / acts:** goal, history, AXTree in; BrowserGym bid-grounded actions out (click, type, scroll, goto).
- **Select:** greedy fixed-budget subset of steps; unary goal relevance plus pairwise diversity over states and answers.
- **Signal:** expert actions kept verbatim; reasoning-native models rewrite traces in their own style.



Our redraw (not the paper's figure): the offline curation pipeline, plus the deployed agent loop in the strip.

$$\max_{J:\,|J|=T_0}\ \sum_{j\in J}\Phi(j)\;+\;\lambda\sum_{i

Importance + λ·diversity under a step budget

Key results: zero-shot online SR (%)

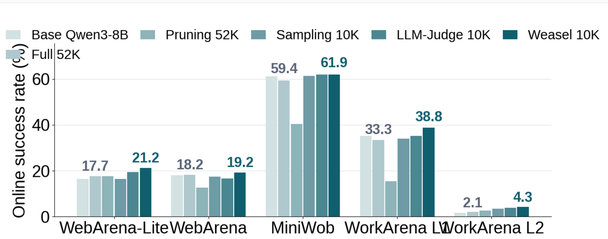
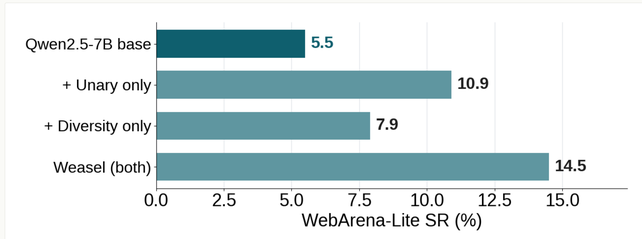


Table 1, redrawn: trained on AgentTrek, Qwen3-8B rows; all online SR in AgentLab.

19% of the data, better than full SFT; 10.7x speed-up

SO WHAT → *Weasel keeps MiniWob at 61.9 (Full 59.4) and lifts WorkArena L2 to 4.3 (base 1.7). Pruning alone is not enough: 40.3 MiniWob.*

Ablation: selection objective (WebArena-Lite SR)



Tables 5-7, redrawn: Qwen2.5-7B, 10K steps, 4 epochs, WebArena-Lite; each bar changes one term of the objective.

SO WHAT → *Both terms matter and compose: unary alone 10.9, diversity alone 7.9, combined 14.5. State-only diversity 9.7, answer-only 13.9; goal-state importance beats goal-summary 12.7 and goal-reason 9.1. On Qwen3-8B, selection and reasoning synthesis are complementary too: without RS 17.0, random+RS 18.2, both 21.2. Hyperparameters are stable: lambda 2.0 gives 12.1, w=80 gives 12.8, T0=0.25T gives 12.1, all under the 14.5 default.*

Headline Numbers

21.2%
WebArena-Lite SR
Qwen3-8B

+4.8 vs Qwen3-8B base

9.7-12.5x training speed-up

19% of full data used

The paper's own controls

CONTROL (QWEN2.5-7B)	RESULT	VERSUS
Uniform 10K sample	9.1	Weasel 14.5
LLM-judge 10K	8.5	under uniform 9.1
Pruning baselines	8.5 / 9.1 / 7.3	ours 10.9
AITW offline (VL-3B)	6.6	random 5.8 / base 4.4

Takeaway

Select, prune, self-explain: 10K curated steps beat full 52K-step SFT zero-shot at 9.7-12.5x lower cost.

7.7 h one-time prep · greedy optimal >96% · NNetNav: 12.1 vs 10.9 · AITW transfer 6.6 · code released

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Paper · Code